



Experimental and kinetic investigation of pyrolysis and oxidation of nitromethane

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ABSTRACT

The pyrolysis and oxidation of nitromethane (NM) were studied in a plug flow reactor over the temperature range of 735–1476 K at 5 Torr and in a jet-stirred reactor over the temperature range of 600–875 K at atmospheric pressure, respectively. Mole fraction profiles of major products and intermediates were identified with tunable synchrotron vacuum ultraviolet photoionization and molecular-beam mass spectrometry. The current study represents the first comprehensive characterization of NM conversion under these conditions. The experimental results were compared to predictions with a detailed chemical kinetic model involving 364 species and 2389 reactions. The rate constants for significant dissociation reactions $\text{CH}_3\text{NO}_2 = \text{CH}_3 + \text{NO}_2$ and $\text{CH}_3\text{NO}_2 = \text{HCNO} + \text{H}_2\text{O}$ were calculated by the RRKM/Master Equation method using Variflex program in this work. The model provided an overall reasonable agreement with the measured data, but for pyrolysis conditions future work is required to improve predictions of some intermediates. Rate-of-production analysis indicates that the primary decomposition pathways of NM are different for pyrolysis and oxidation due to the effects of C–N bond fission and roaming mediated isomerization. From the sensitivity analysis, the two pathways mentioned above have promoting effects on NM consumption for all the studied conditions, while the reaction of $\text{CH}_3 + \text{NO}_2 = \text{CH}_3\text{O} + \text{NO}$ with inhibiting effect at $\Phi = 2.0$ shows promoting effects for $\Phi = 0.4$ and pyrolysis. This work extends the experimental database and helps to improve the understanding of low temperature chemistry of NM.

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1. Introduction

Nitromethane (NM), the simplest organic nitro compound, is a commonly used solvent for industrial applications, a fuel regularly used in model engines or race engines, and an additive to fuel for internal combustion engines [1,2]. Meanwhile, NM is also used as a model compound to simulate monopropellant combustion [3] due to its dual fuel-oxidizer characteristics. Therefore, it is desired to investigate the combustion characteristics of NM, serving also to obtain a better understanding of fuel-N chemistry.

Previous studies mainly focused on the detonation and monopropellants characteristics of NM [4–7]. Only a few experimental and kinetic studies of NM combustion characteristics have been reported [1,2,8–20]. In 1999, Glarborg et al. developed a kinetic

model of NM decomposition based on reported experiments covering temperatures of 1000–1400 K and pressures of 0.5–6.0 atm [14]. Later, the detailed structure of premixed flames of NM/O₂/Ar at low pressure were reported by Tian et al. [15] and Zhang et al. [16]. The model from Zhang et al. was adopted with a few changes by Brequigny et al. [1]. Nauc  r et al. [19] investigated the ignition of NM/O₂/N₂ mixtures via shock tube experiments in the temperature range 947–1333 K at reflected shock pressures near 8, 16 and 32 atm. The laminar burning velocities of NM/air flames at 1 atm and initial gas temperatures of 338 K, 348 K, and 358 K were studied in non-stretched flat flames by Nauc  r et al. [20]. The most recently published mechanism was assembled by Mathieu et al. [17] based on their work on ignition delay times for NM/O₂/Ar at 2–35 atm and 875–1595 K. The experimental study on premixed flames of NM at atmospheric pressure was performed by Brackmann et al. [18]. Roaming mediated isomerization of NM was studied by Annesley et al. [21]. However, the above mechanisms were all developed based on high temperature or/and high pressure

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data; none of them has been validated against experimental data of NM pyrolysis and oxidation at low to intermediate temperature. Therefore, to further validate NM mechanisms over a wider range of conditions, additional experimental data, especially the chemical structures and concentration profiles, are desired.

This work aims to investigate the pyrolysis and oxidation of NM. Pyrolysis experiments were carried out over the temperature range of 735–1476 K at 5 Torr in a plug flow reactor, and oxidation experiments were performed at 600–875 K and 1 atm at fuel-air equivalence ratios Φ of 0.4 and 2.0 using a Jet-stirred reactor (JSR). Concentrations of major products and intermediates were identified with tunable synchrotron vacuum ultraviolet photoionization and molecular-beam mass spectrometry (SUUV-PI-MBMS). Based mainly on previous studies [16,21,22], a detailed chemical kinetic mechanism of NM was proposed. The work extends the experimental database for NM pyrolysis and oxidation, and provides a better understanding of fuel-N chemistry.

2. Experimental

The experiments were performed at National Synchrotron Radiation Laboratory (NSRL), Hefei, China. A detailed description of the instruments can be found elsewhere [23–26].

For NM pyrolysis, the details of pyrolysis experiment setup can be found elsewhere [26]. In brief, an electrically heated flow reactor was placed in the center of a chamber named pyrolysis chamber. The pressure was kept at 5 Torr. NM (0.063 mL/min, liquid phase) was vaporized at 400 K by heating and diluted with 95% of argon (0.50 standard liter per minute, 99.999% by volume, gas phase). The mixture was fed into a 220 mm long Al_2O_3 flow tube with an inner diameter of 6.8 mm. The temperature was measured along the central line of the tube. For each temperature profile, the corresponding pressure distribution was calculated. A detailed description of the temperature measurement and pressure calculation can be found elsewhere [24]. The temperature and pressure profiles are provided in the Supplementary Material (SM).

For the oxidation experiments, the JSR used in this work consisted of a small sphere with a volume of 103 cm^3 . NM was evaporated in a gasification chamber at 400 K, above the boiling point of 373 K. The NM oxidation was investigated at 1 atm and fuel-air equivalence ratios of $\Phi = 0.4$ and 2.0, considering CO_2 , NO and H_2O as major products. The residence time was kept at 2.0 s at both conditions, with a temperature range of 600–875 K. The initial mole fractions of NM were kept at 1% for both conditions, and those of Ar and O_2 are 95.875% and 3.125% for $\Phi = 0.4$, and 98.375% and 0.625% for $\Phi = 2.0$.

The gas products were identified by comparing the measured ionization thresholds to literature values. Some photon energies were selected to avoid fragmentation of intermediates and distinguish isomeric intermediates. The photoionization cross-sections of most species identified are available in the literatures; however, for some nitrogenous intermediates (such as HCNO, HNCO, HONO, and HNO) few data are available. Therefore, an estimated value of 10 Mb (a unit of cross section, 1 Mb = $1\text{E}-18 \text{ cm}^2$) was used when the photon energy was 0–0.95 eV higher than the IEs; and 20 Mb was used when the photon energy was 1–2 eV higher than the IEs [15]. The photoionization cross sections at different photon energies used in this work is provided in SM. The uncertainties of the mole fractions are ± 5 –10% for the major species, ± 25 % for intermediates with known photoionization cross sections, and a factor of two for those with estimated photoionization cross sections.

3. Modeling

The modeling for NM oxidation was performed using the Perfectly Stirred Reactor (PSR) code from Chemkin-II software [27],

and modeling for NM pyrolysis used Plug Flow Reactor (PFR) code from the Chemkin-PRO software [28]. The kinetic scheme established here is coupled with our recent mechanism which has been developed for the oxidation of acetylene [22], by adding a sub-mechanism for NM [16]. Roaming mediated isomerization of NM studied by Annesley et al. [21] is considered in this work. We have also studied theoretically the kinetics of the primary decomposition of NM by using ab initio simulations and master equation analysis. The potential energy surface was provided in Fig. 1. Geometry optimization and rovibrational frequencies of all the stationary points were calculated with the B3LYP/6-311++G(3df,2p) level of theory [29–33]. To achieve more accurate energetics, the CCSD(T)/6-311+G(3df,2p) [34] method was used to perform the single-point energy calculations. Note that we have only located a tight transition state for the isomerization $\text{CH}_3\text{NO}_2 \rightarrow \text{CH}_3\text{ONO}$, where the CH_3ONO is a trans. The loose transition state, which leads to a cis conformer, was not located at the current level. All quantum chemistry calculations were performed with the Gaussian 03 program [35].

The rate constants for significant dissociation reactions $\text{CH}_3\text{NO}_2 = \text{CH}_3 + \text{NO}_2$ and $\text{CH}_3\text{NO}_2 = \text{HCNO} + \text{H}_2\text{O}$ were calculated by the RRKM/Master Equation method using Variflex program [36], and comparisons of rate constants of above two reactions and the roaming isomerization reaction from Annesley et al. [21] are provided in Fig. 2. It is shown that the rate constants of reactions $\text{CH}_3\text{NO}_2 = \text{CH}_3 + \text{NO}_2$ and $\text{CH}_3\text{NO}_2 = \text{CH}_3\text{O} + \text{NO}$ are similar at low temperature, and rate constant of the reaction of C–N bond cleavage is higher than that of roaming isomerization at high temperature for 760 Torr. It is obvious that the rate constants of above two reactions are higher than that of reaction $\text{CH}_3\text{NO}_2 = \text{HCNO} + \text{H}_2\text{O}$. The difference indicates that the cleavage of C–N bond and roaming isomerization are dominant pathways of NM while the effect reaction of $\text{CH}_3\text{NO}_2 = \text{HCNO} + \text{H}_2\text{O}$ on NM consumption can be ignored, which is consistent with the main conclusions discussed later in this work.

The complete mechanism involves 364 species and 2389 reactions in SM. Rate-of-production (ROP) and sensitivity analyses were performed to track the key reaction pathways governing NM consumption and intermediates production. Besides the present mechanism, predictions with the models from Zhang et al. [16], Brequigny et al. [1], and Mathieu et al. [17] were also compared with the experimental results; these are shown in SM.

4. Results and discussion

4.1. New intermediates

The photoionization efficiency spectra (PIE) of newly observed intermediates during NM oxidation are provided in Fig. 3. By scanning PIE, major products and intermediates were both identified in the pyrolysis and oxidation processes. All the pyrolysis intermediates have also been reported from NM flame studies [15,16], while methoxy radical, formoxime, methyl hydroperoxide, and acetaldoxime were newly detected in the oxidation of NM in this work. A complete species list is available in SM. For $m/z = 31$, the onsets at 9.40, 10.10 and 10.75 eV correspond to the ionizations of methanamine (IE = 9.45 eV [37]), nitrosyl hydride (IE = 10.10 eV [37]), and methoxy radical (IE = 10.72 eV [37]). The $m/z = 45$ was assigned to formoxime (IE = 10.11 eV [37]) and nitrosomethane (IE = 10.80 eV [37]). For the other two curves in Fig. 3(c) and d, PIE spectra identify these species to be methyl hydroperoxide ($m/z = 48$, IE = 9.91 eV [38]) and acetaldoxime ($m/z = 59$, IE = 10.70 eV [37]). These new intermediates can provide more detailed kinetic information of NM. The above newly observed intermediates and their derived species (such as HCN and CH_2NO from formoxime, CH_3O , CH_2OOH , and CH_3O_2 from methyl hydroperoxide, CH_3CN and CH_3CHNO from

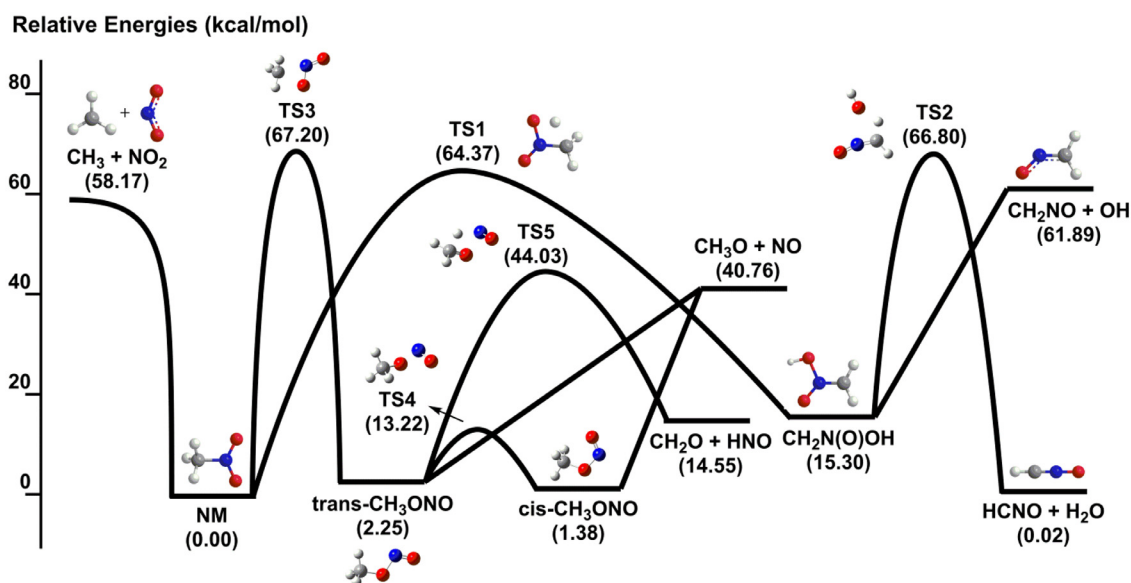


Fig. 1. Potential energy surface for nitromethane decomposition in kcal/mol at 0 K, computed at CCSD(T)/6-311+G(3df,2p)//B3LYP/6-311++G(3df,2p).

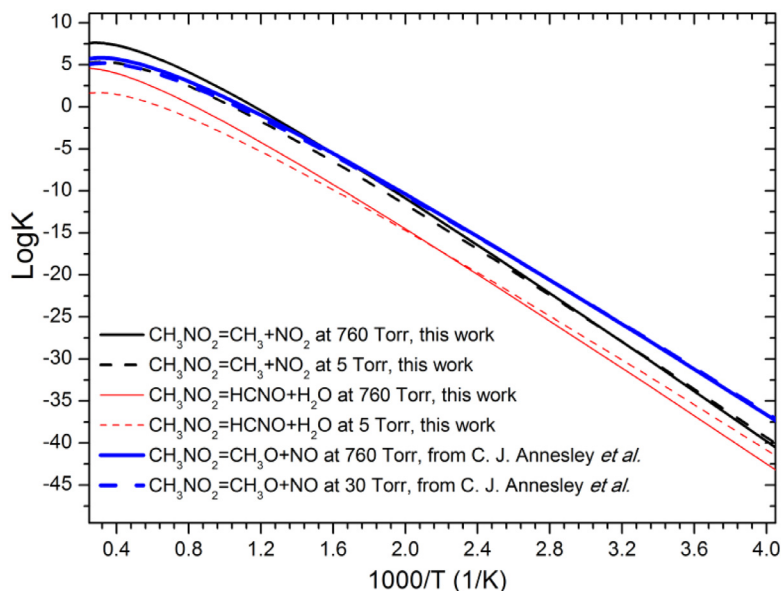


Fig. 2. Rate constants comparison of reactions $\text{CH}_3\text{NO}_2 = \text{CH}_3 + \text{NO}_2$ and $\text{CH}_3\text{NO}_2 = \text{HCNO} + \text{H}_2\text{O}$ at 760 Torr and 5 Torr, as well as the reactions of $\text{CH}_3\text{NO}_2 = \text{CH}_3\text{O} + \text{NO}$ from Annesley et al. [21].

acetaldoxime) [39] were considered in the model, but they have no effect on major products.

4.2. Mole fraction profiles

The experimental and predicted mole fraction profiles of pyrolysis and oxidation species are shown in Figs. 4–8. The pyrolysis results for major species and intermediates are shown in Figs. 4 and 5. The decomposition of NM starts between 950 K and 1000 K under the conditions used in this work, which is higher than that of NM oxidation. NO_2 and CH_3 are detected as the primary products of the C–N bond fission (Figs. 4(d) and 5(a)). NO, the major nitrogenous product, appears at higher temperatures, formed mainly from $\text{CH}_3 + \text{NO}_2 = \text{CH}_3\text{O} + \text{NO}$ and $\text{CH}_3\text{NO}_2 = \text{CH}_3\text{O} + \text{NO}$. The subsequent decomposition of CH_3O yields CH_2O , based on the ROP analysis, which will be discussed in detail in next section.

From Fig. 5, it is seen that some of the pyrolysis intermediates are not predicted well by the model. In particular, the formation of HNCO and HCNO is severely underpredicted. HCNO appears at temperatures lower than that of HNCO. The formation of HNCO was proposed to be mainly through the conversion of $\text{NO} \rightarrow \text{HCN} \rightarrow \text{HNC} \rightarrow \text{HNCO}$ in flames [16], in accordance with its appearance at relatively higher temperatures in the present work. However, for HCNO (Fig. 5(d)) the peak occurs earlier and is much higher than predicted. According to the model, the formation of HCNO is mainly from the reaction between NO and the products from the dehydrogenation and oxidation of hydrocarbons [15,16], i.e., $\text{NO} + \text{CH}_2$ and $\text{NO} + \text{HCCO}$. However, these steps are not significant at the conditions of the present work, since the CH_2 and HCCO radical levels are negligible at low temperatures. HCNO is firstly detected at the temperature that NM starts to decompose, when the yield of NO_2 and CH_3 are still limited. As expected, the concentrations of NO and CH_2O at this temperature are near-zero,

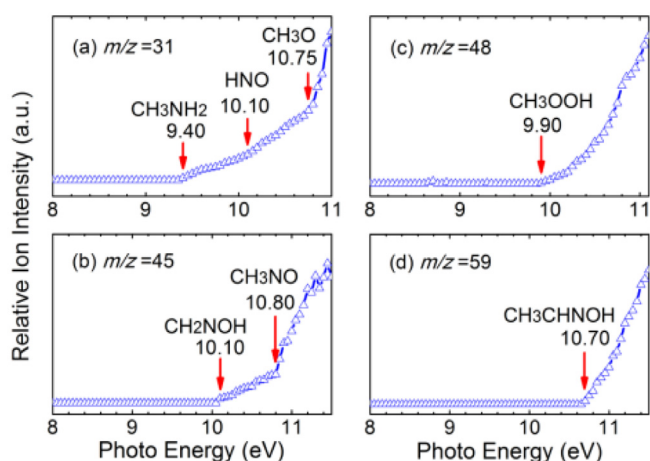


Fig. 3. Photoionization efficiency spectra of $m/z = 31, 45, 48$ and 59 sampled in the oxidation of NM at temperature of 825 K and $\Phi = 0.4$.

since these species are formed from secondary reactions. As the temperature increases, the mole fraction of HCNO keeps increasing, from which it can be concluded that the signal of HCNO is not from a fragment of NM produced in the photoionization process. The data indicate that HCNO is among the primary decomposition products of NM, rather than a secondary product from the reaction of CH_2 and NO. Such a reaction of giving HCNO from NM is considered in present model, but the predicted level of HCNO is much lower than observed. However, the discrepancy for HCNO and HNCO between experimental data and prediction may partly be attributed to the estimated photoionization cross sections used in the quantification process. The pathway of NM decomposition forming HCNO has to be re-estimated in future work.

The major species of NM oxidation are shown in Fig. 6. Figures 6(a) and (b) present the mole fractions of reactants. For both values of Φ , the decomposition of NM was observed to occur at a temperature above 725 K , and the concentration profiles were almost the same. The mole fractions of O_2 and Ar decreased within the same temperature region as that of NM consumption for both Φ . The experimental mole fraction of CO at $\Phi = 0.4$ was higher than that of CO at $\Phi = 2.0$ below 850 K , but above this temperature the contrary trend was observed. As expected, the CO_2 mole fraction at lean conditions was higher than that at rich conditions, increasing with temperature for both cases. NO was detected as the major nitrogenous product, indicating that NO was more stable than NO_2 in NM oxidation at low temperatures, similar to the observations in an NM flame [16] and under pyrolysis

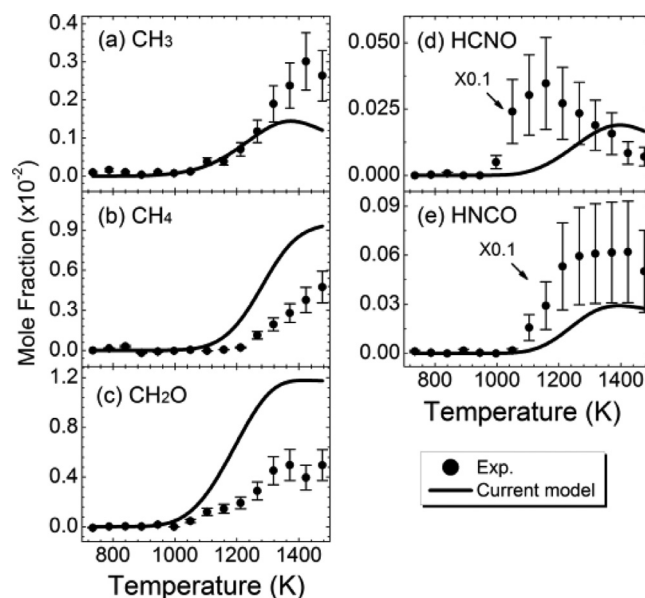


Fig. 5. Mole fraction profiles of intermediates in NM pyrolysis.

conditions. However, N_2 and NO were found to be the major nitrogenous products in NM flame [15]. Figure 6(g) shows that the mole fraction profiles of H_2O have similar tendency as those of CO_2 at both Φ . NO_2 is reactive at high temperature, and therefore identified as an intermediate with the maximum concentrations at the temperature of 800 and 775 K for $\Phi = 0.4$ and 2.0 , respectively. Modeling results of major species were also exhibited in Fig. 6. In general, better agreement is observed between experimental and predictions by the present model than those from Zhang et al. [16], Brequigny et al. [1], and Mathieu et al. [17] (see SM).

The experimental and predicted mole fractions of hydrocarbon intermediates in NM oxidation are shown in Fig. 7. The mole fractions of CH_4 , C_2H_2 , C_2H_4 , C_2H_6 and CH_3OH increase with increasing temperature, attaining the highest concentrations for $\Phi = 2.0$. The most abundant hydrocarbon intermediate is CH_4 , formed from CH_3 , after the C–N bond rupture of NM. The present mechanism tends to underpredict the peak values of some hydrocarbon intermediates, probably due to the uncertainty within the elementary reactions and unrevealed pathways. CH_2O reaches peak values at 800 and 875 K for $\Phi = 0.4$ and 2.0 and was well simulated.

Figure 8 summarizes the results for nitrogenous intermediates in NM oxidation. As Φ increases, the mole fractions of HCN, N_2O , N_2 , HNO and HNCO become higher at the same

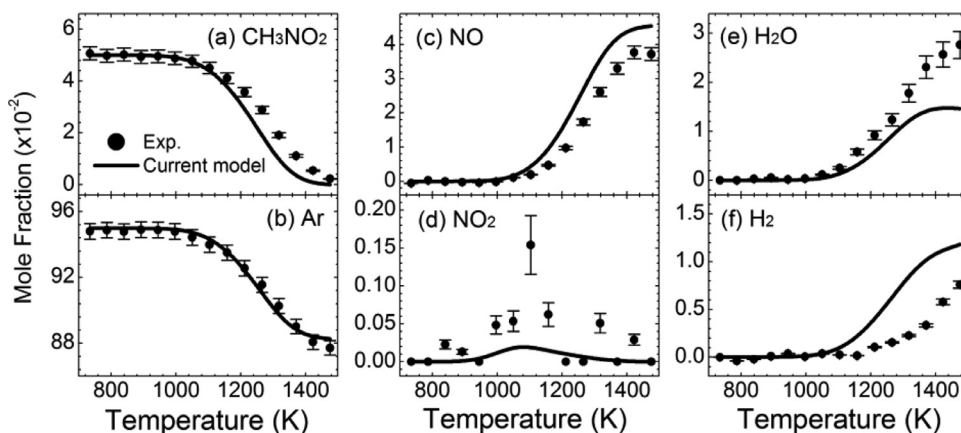


Fig. 4. Mole fraction profiles of major species in NM pyrolysis.

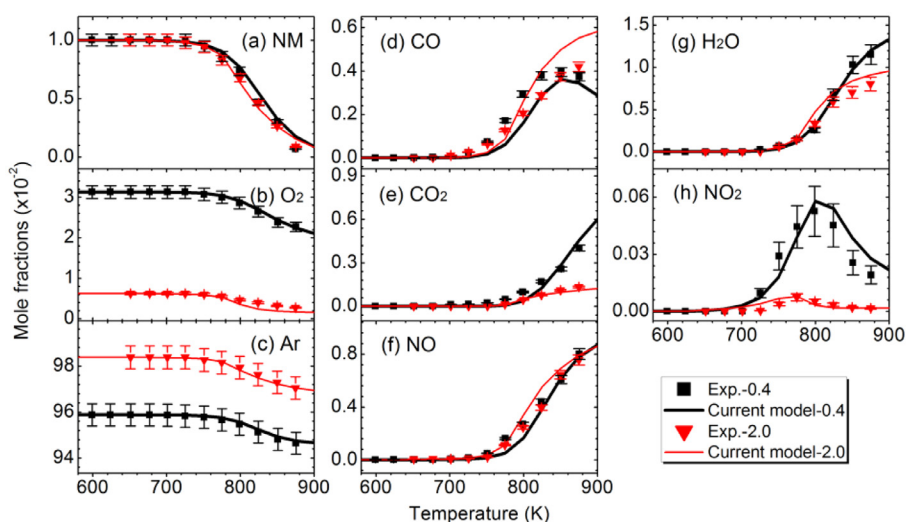


Fig. 6. Mole fraction profiles of major species in NM oxidation.

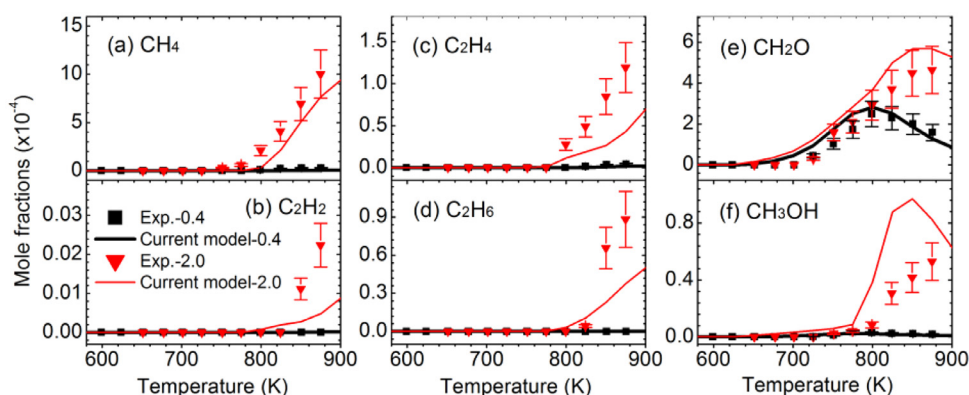


Fig. 7. Mole fraction profiles of selected hydrocarbon species in NM oxidation.

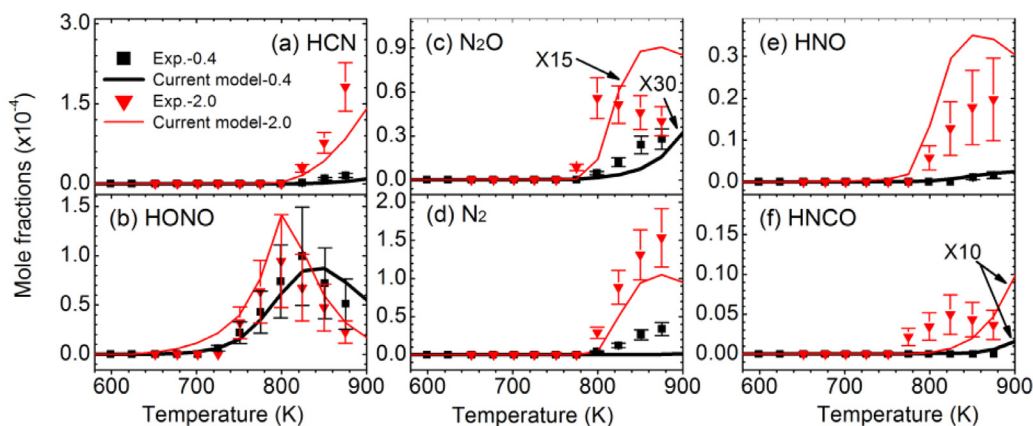


Fig. 8. Mole fraction profiles of nitrogenous intermediates in NM oxidation.

temperature. Based on ROP analysis, nitrogenous intermediates are formed via the sequences of $\text{NO} \rightarrow \text{HNO} \rightarrow \text{HONO}/\text{N}_2\text{O}/\text{N}_2$ and $\text{CH}_3 + \text{NO} \rightarrow \text{HCN} \rightarrow \text{HNC} \rightarrow \text{HNCO}$. The most abundant nitrogenous intermediate is HCN. The concentrations of HONO at different Φ and the same temperature are similar, and at both Φ it is well predicted. The present mechanism underpredicts the concentrations of N_2O and HNCO. N_2O is mainly formed from HNO. As HNO is a little overpredicted, the deviation for N_2O may result from the uncer-

tainty within related reactions. Similarly, the discrepancy for HNCO could be due to the underestimation of HCN. Compared to NM flame [15], NO is detected as the only major nitrogenous product in NM oxidation. N_2 mainly comes from the oxidation sequence of $\text{NO} \rightarrow \text{HNO} \rightarrow \text{H}_2\text{NO} \rightarrow \text{NH}_2 \rightarrow \text{N}_2$, and the pathway of NO giving HNO accounts for about 39.4% and 66.4% of NO consumption at $\Phi = 0.4$ and 2.0, respectively, which leads to the lower concentration of N_2 compared to NO in oxidation.

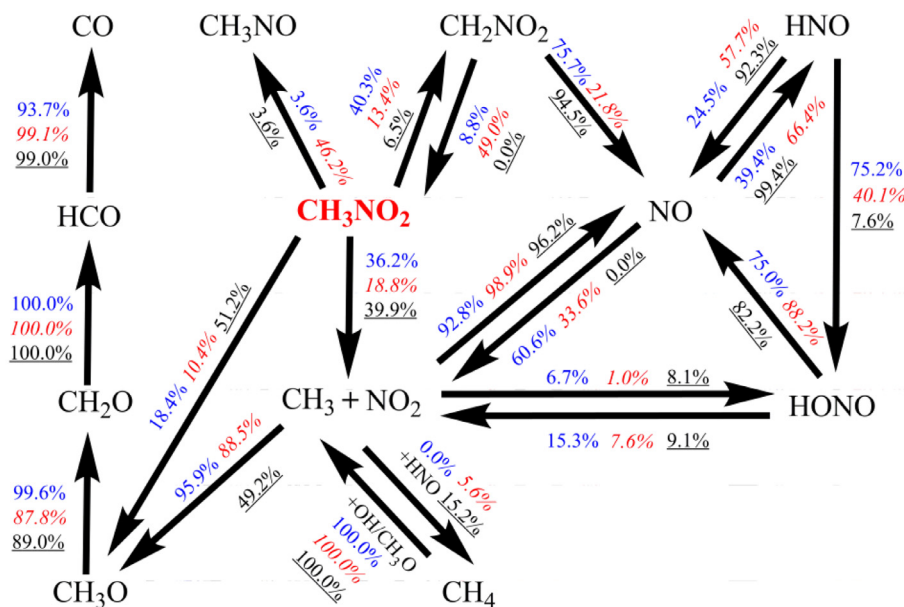


Fig. 9. ROP analysis of NM pyrolysis (underline and black) and oxidation under $\Phi = 0.4$ (blue), 2.0 (italic and red), corresponding to 1300 K, 56% NM conversion, 825 K, 54% NM conversion, and 825 K, 54% NM conversion, respectively. The percentages are integrated contributions to species consumption. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

4.3. ROP analysis

To illustrate the major reaction channels of NM pyrolysis and oxidation, a flux analysis has been performed for pyrolysis (1300 K, 56% NM conversion) and oxidation at $\Phi = 0.4$ (825 K, 54% NM conversion) as well as 2.0 (825 K, 54% NM conversion). The results are shown in Fig. 9.

For pyrolysis, NM exhibits similar pathways to oxidation process. However, the ratios of some reactions are dramatically different. It is noticed that the dominant consumption pathway for NM is the roaming mediated isomerization of NM, which accounts for 51.2%, significantly higher than in the cases of oxidation. This is consistent with the conclusion drawn by Annesley et al. [21] that roaming pathway is dominated in NM decomposition when the pressure is below 200 Torr. Besides, the cleavage of C–N bond is the second major reaction for NM pyrolysis. In the absence of O_2 , formation of partly oxygenated species such as CH_3O , CH_2NO_2 , and HONO is less important in NM pyrolysis, while the formation of CH_4 and NO/HNO recycling becomes more pronounced. Although reactants and products of some reaction are the same in NM pyrolysis and oxidation, the overall reaction paths are different. For example, HNO is recycled to NO mainly by reaction with NO_2 and O_2 in NM oxidation, and by the reactions with CH_3 and CH_3O during pyrolysis. The difference between NM pyrolysis and oxidation results from differences in oxygen content, temperature, pressure, and reactor configuration. Except NM roaming pathway, the ROP analysis of NM flames [16] shares similarities with both pyrolysis and oxidation processes studied in the present work.

For oxidation, it is noticed that the equivalence ratio has obvious effect on NM consumption under the investigated conditions. At the lean condition, the major consumption pathway for NM is $CH_3NO_2 + OH = CH_2NO_2 + H_2O$, which accounts for 40.3%, and the reaction of the cleavage of C–N bond accounts for 36.2%. At rich conditions, the reaction of $CH_3NO_2 + CH_3 = CH_3NO + CH_3O$ accounts for 46.2% on NM consumption, and the cleavage of C–N bond only accounts for 18.8%. Almost all CH_3NO would then produce CH_3 and NO via the cleavage of C–N bond. For both Φ , the contributions of roaming mediated isomerization on NM consumption are less than 20%. When Φ increases, the concentration of CH_3 increases while

OH decreases, leading to the difference in the major consumption pathways for NM oxidation. The CH_3 radical is mostly converted to CH_3O , which then mainly forms CH_2O . Through the reaction with OH, H, and CH_3 radicals, all CH_2O forms HCO, which subsequently yields CO via reaction with NO, O_2 and H. The CH_3 radical also reacts to form minor amounts of CH_4 at rich conditions, but this was not observed for lean conditions. Conversion of NO_2 formed by the initial decomposition of NM may lead to a number of nitrogenous intermediates. The most important consumption pathways of NO_2 involve hydrocarbons. The reactions of NO_2 with CH_3 and H producing NO account for 92.8% and 98.9% of the NO_2 consumption at $\Phi = 0.4$ and 2.0. Besides, some NO_2 is converted to HONO, which then mainly forms NO. NO can also be produced from the CH_3NO and CH_2NO_2 radicals. There is a significant recycling between NO, HNO, HONO and NO_2 for both Φ .

4.4. Sensitivity analysis

To identify the reactions that serve as bottlenecks in the consumption of NM, a sensitivity analysis has been performed with the present mechanism both for oxidation and pyrolysis, as shown in Fig. 10. The reactions of $CH_3NO_2 = CH_3O + NO$, $CH_3NO_2 = CH_3 + NO_2$, and $CH_3NO_2 + CH_3 = CH_3NO + CH_3O$ are the most important promoting steps for NM conversion at pyrolysis, lean and rich conditions, respectively. The result indicates that the formation and consumption of CH_3 , NO, and NO_2 play key roles for NM pyrolysis and oxidation. The reaction $CH_3 + NO_2 = CH_3O + NO$ is the most sensitive terminating step for oxidation at $\Phi = 2.0$, while it promotes conversion at $\Phi = 0.4$ and pyrolysis. The most sensitive terminating steps for oxidation at $\Phi = 0.4$ and pyrolysis are $HONO + OH = NO_2 + H_2O$ and $CH_3NO_2 + H = CH_2NO_2 + H_2$, respectively.

4.5. Validation against other experimental data

To check and to extend the validity range of the current model, previous ignition delay times measurements from Mathieu et al. [17] (at $p = 2.0$ atm with NM diluted in Ar, $\Phi = 1.0$) and Nauclér et al. [19] (at $p = 8.0$ atm with NM diluted in N_2 , $\Phi = 0.5$) were

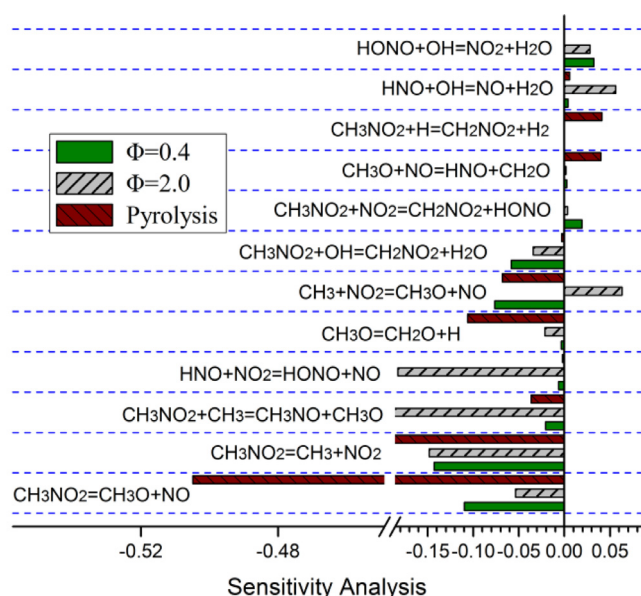


Fig. 10. Sensitivity analysis of NM pyrolysis and oxidation under $\Phi = 0.4, 2.0$, corresponding to 1300 K, 56% NM conversion, 825 K, 54% NM conversion, and 825 K, 54% NM conversion, respectively.

simulated. Both ignition delay times were studied in high-pressure shock tubes, respectively [17,19].

Figure 11 shows comparison between the measured and simulated data by using the current model and other mechanisms from Zhang et al. [16], Brequigny et al. [1], and Mathieu et al. [17]. Under the both conditions, the ignition delay times decrease with temperature increasing. The tendency of the simulated results by current model shows a good agreement with the experimental data. From Fig. 11(b), ignition delay times from current model are slightly lower than the measurements, while show better prediction results than other mechanisms. This underestimation is probably mainly caused by the lower consumption speed of NM in shock tubes at high pressure than that of current model. In summary, the current model has reasonable agreements on the ignition delay time measurements of NM.

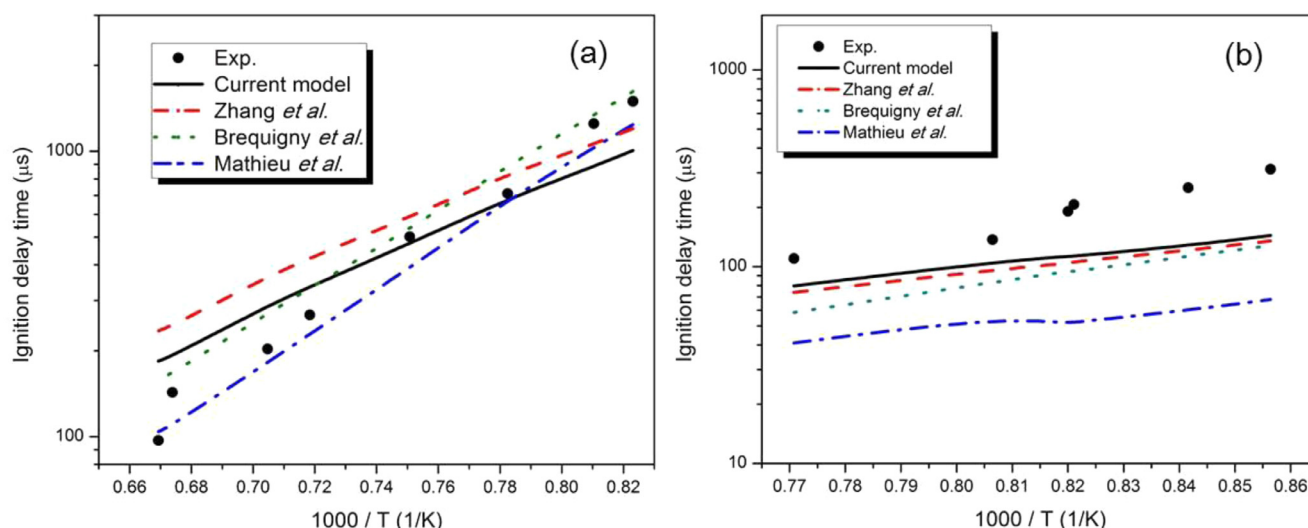


Fig. 11. Comparison between the predicted (curves) and measured (symbols) ignition delay time profiles: experimental data - Mathieu et al. [17] (left part) and Nauc  r et al. [19] (right part); modeling: current model compared to other mechanisms [1,16,17].

5. Conclusion

This work reports an experimental and kinetic investigation of NM pyrolysis and oxidation. The mole fractions of products were identified by SUVU-PI-MBMS, including some new observed intermediates, such as methoxy radical, formoxime, methyl hydroperoxide, and acetaldoxime during oxidation process. Based on the measurements and calculated results, a detailed chemical kinetic model was established. NO is found to be major nitrogenous product for pyrolysis and oxidation, which is different from the NM flame. In this work, the equivalence ratio has obvious effect on main consumption pathway of NM due to ratios in CH_3 and OH radical concentrations. The ROP analysis indicates that the NM consumption is dominated by roaming mediated isomerization ($\text{CH}_3\text{NO}_2 = \text{CH}_3\text{O} + \text{NO}$) for pyrolysis, and the reactions of $\text{CH}_3\text{NO}_2 + \text{OH} = \text{CH}_2\text{NO}_2 + \text{H}_2\text{O}$ and $\text{CH}_3\text{NO}_2 + \text{CH}_3 = \text{CH}_3\text{NO} + \text{CH}_3\text{O}$ are main channels for NM consumption at lean and rich conditions, respectively. According to the sensitivity analysis, roaming mediated isomerization is the most important promoting steps for pyrolysis, the reactions of C–N bond cleavage and $\text{CH}_3\text{NO}_2 + \text{CH}_3$ are most sensitive for oxidation consumption at lean and rich conditions. Further studies are required to understand the role of HCNO in the primary decomposition of NM and to refine relevant reactions involving HCNO. This work will benefit an improved understanding of low temperature fuel–N chemistry.

Acknowledgments

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Supplementary material

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